

Scene contrast temperature in two different climates

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Scene contrast temperature (SCT) is a measurement of temperature and photon flux variation in an emissive scene used to calculate targeting task performance, a metric used to determine the ability of an observer to perform visual tasks. While previous research has compared the SCT of midwave infrared (MWIR) and longwave infrared (LWIR) systems in a hot, dry climate, little work has been done to compare the performance of these systems in different climates. This paper compares the SCT of a hot, dry climate and a cold, dry climate in MWIR and LWIR. The results indicate that SCT is higher in the LWIR in most conditions, except for cold, daylight environments. © 2025 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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1. INTRODUCTION

Longwave infrared (LWIR), which spans the wavelengths 8–12 μm , is used for rotorcraft pilotage systems, but there may be benefits to using midwave infrared (MWIR), which spans the wavelengths 3–5 μm , for these systems going forward. MWIR has superior atmospheric transmission and reduced diffraction spot sizes relative to the LWIR but suffers from reduced in-band photons [1]. Additionally, weather conditions like fog and dust affect LWIR and MWIR systems differently. For example, LWIR has superior visibility in fog [2,3], while MWIR has superior visibility when snow is falling [4]. Because of this, which band a pilotage system operates is an important choice for system designers to make.

An important metric when choosing an operating band is the flux variance of sources in the environment. For systems intended to detect and identify targets, low flux variance is advantageous due to the reduction in scene clutter and increased contrast of targets against the background [1,5]. On the other hand, for systems like pilotage and image-based navigation, high flux variance is favorable because the increased variance adds to the user's situational awareness [1,5]. Figure 1(A) shows an image with high variance where features, such as the cacti, shrubs, and hills, are easily recognizable, while Fig. 1(B) shows an image with low variance, where it is difficult to see the same features.

This paper presents the flux variance measured in urban and rural environments, during the day and night, in both hot and cold climates. Additional measurements are presented from

a snowy scene and a rainy scene. Diurnal measurements are presented from both locations.

2. BACKGROUND

A purely emissive scene is comprised of a background with a relatively large flux, and the objects present with fluxes that deviate, often minimally, from the background [1,6]. These objects generate a variance in flux across the scene. The degree of variance is affected by the number of objects present and the difference between the object and background flux. Objects are discriminated against the background by their emissive contrast, which is a combination of their temperature difference and emissivity difference, and thus, temperature variance is an important metric when determining the best system for a given task.

One measure of global radiometric variance, or variance across the entire image, is scene contrast temperature (SCT). SCT is defined as the change in radiometric temperature needed to generate average luminance when the minimum luminance is zero [7]. A simpler definition of SCT is that it is the difference between the background temperature and the average temperature [5,7]. A measure of local variance, or variance within a subregion, is the standard deviation. This measure is beneficial because it allows for scenes with different fields of view (FOVs) to be compared. The variance of a scene can be affected by any factor that changes its thermal or lighting conditions such as overcast skies, rain, or humidity. The variance of a scene is also affected by the emissivity of the objects present.

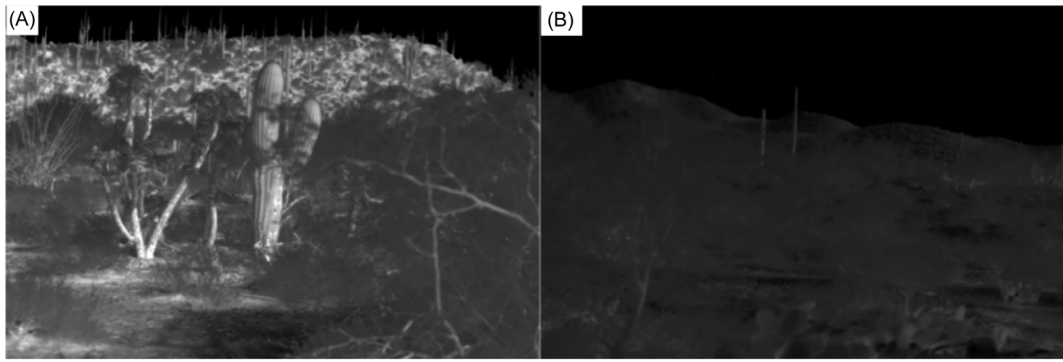


Fig. 1. Two LWIR images taken in a rural setting. Image A has a higher variance with a standard deviation of $\sim 3.89^\circ\text{K}$ than Image B, which has a standard deviation of $\sim 0.79^\circ\text{K}$.

The effect of SCT on system performance is mathematically quantified by Vollmerhausen and Jacobs [7]. Equation (1) provides the relationship between target contrast (C_{tgt}) and SCT:

$$C_{\text{tgt}} = \frac{T_{\text{tgt}}}{2\text{SCT}}, \quad (1)$$

where T_{tgt} is the target's blackbody equivalent temperature.

The TTP metric, used in models like the U.S. Army DEVCOM's night vision integrated performance model (NV-IPM) to calculate the probability of visual task performance from scene and system parameters [8], is determined by C_{tgt} and the contrast threshold function (CTF) of the observer [1,7]:

$$\text{TTP} = \int_{\xi_{\text{low}}}^{\xi_{\text{cut}}} \sqrt{\frac{C_{\text{tgt}}}{\text{CTF}}} d\xi, \quad (2)$$

where ξ_{low} is the lowest frequency observable by the system, and ξ_{cut} is the cutoff frequency. The CTF is a function that describes the contrast required for an observer to perceive a given spatial frequency. Since C_{tgt} is dependent on SCT, the SCT has a direct impact on the ability of an observer to perform visual tasks.

3. EXPERIMENT

A. Sensors

We used an FLIR T1020sc to collect measurements for the LWIR and a Telops Spark-M150 with a $3\text{ }\mu\text{m}$ longpass filter to collect measurements for the MWIR. Table 1 provides the sensor specifications. Note that the FOV of the LWIR camera was wider than that of the MWIR, and thus, a global variable like SCT would have observed different scenes for each camera. This was corrected by analyzing the variance of a subregion that is present in both scenes using the standard deviation.

B. Calibration

Calibration was performed using a single blackbody. The temperature of the blackbody started at 20°C , then was increased in steps of either 10°C or 5°C to a final temperature of 50°C . The calibration measurements were then imported into Python as RAW 16-bit single-channel data for the LWIR and MWIR cameras. Then, an ROI was chosen to encompass the surface of the blackbody. After this, the average of the gray levels observed

Table 1. Sensors Used and Their Technical Data

	MWIR	LWIR
Name	Telops Spark-M150	FLIR T1020sc
Detector type	Cooled InSb	Uncooled microbolometer
Spectral range (μm)	3.0–5.4	7.5–14
Spatial resolution	640×512	1024×768
Pitch (μm)	15	17
Optical aperture	f/3	f/1.1
NETD (mK)	20	20
Lens focal length (mm)	25	36
Horizontal FOV	21.7°	27.2°
Vertical FOV	17.5°	20.6°

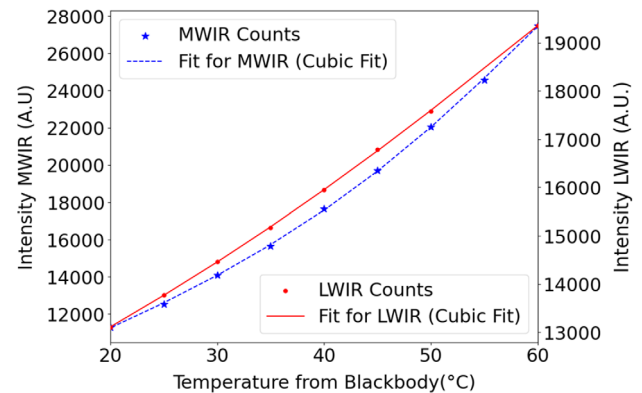


Fig. 2. SITF for the LWIR (T1020SC) and MWIR (Spark M150) detectors. The temperature was increased in steps of 5°C here to better display the cubic fit.

in this region was calculated, and a cubic line of best fit was found to relate the value of the pixels observed to the known temperature of the blackbody. The resulting signal intensity transfer function (SITF) is shown in Fig. 2. Calibrations were taken at the beginning of each measurement.

Due to nonlinearities present, a cubic is fit to the SITF. The difference in accuracy between a cubic and linear fit is shown in Fig. 3. A cubic fit was used since it was the lowest-order polynomial that caused the error values in both cameras to display no clear relationship with temperature.

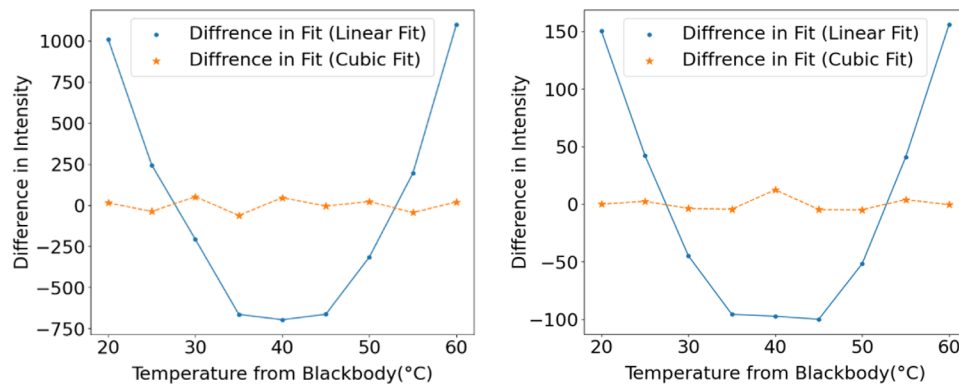


Fig. 3. Difference in intensity observed compared to ideal intensity. Left shows the difference from the MWIR calibration, while the right shows for the LWIR calibration.

Table 2. Date, Time, Location, and Weather Conditions of the Measurements Taken in Arizona

Date	Time	Location	Purpose of Measurement	MWIR Integration		Temperature	Humidity
				Time (μ s)			
October 12	15:30–17:30	Saguaro	Rural day	1000		37.2–38.9°C	7%
October 21	15:00–16:00	Tucson, AZ	Urban day	800		28.8–30°C	8%
October 25	22:00–23:55	Tucson, AZ	Urban night	1500		18.3–20.0°C	16%
October 31	21:00–22:30	Saguaro/Snyder Hill	Rural night	1500		14.4–17.2°C	20%
November 3–4	15:00–18:00	Tucson, AZ	24 h diurnal	1500		6.1–16.1°C	34%–90%

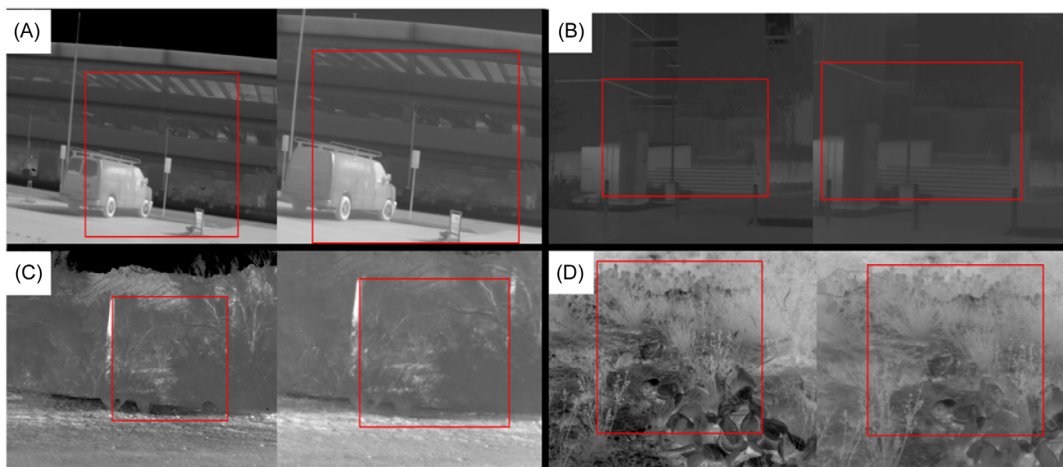


Fig. 4. Measurements taken in Arizona: daytime urban measurements (A); nighttime urban measurements (B); daytime rural measurements (C); and nighttime rural measurements (D). For each subfigure, the image on the left is of the LWIR and the image on the right is of the MWIR scene.

The cubic fit generated from the SITF is then used to map the gray levels observed in each scene to its blackbody equivalent temperature. No additional post-processing is performed.

C. Measurements Taken in Arizona

Measurements were taken in Arizona—a hot, dry climate—during October and early November. Measurements were taken in both urban and rural desert locations. Urban measurements were taken in Tucson, in the vicinity of the University of Arizona’s main campus. The objects present in urban scenes included buildings, cars, roads, power lines, and other common objects. Rural desert measurements were taken in Saguaro

National Park and the Snyder Hill Bureau of Land Management Camping Area. The objects present in rural scenes primarily included scrub brushes, cacti, and rocks. Table 2 lists the conditions during each Arizona measurement, and Fig. 4 shows reference images from the Arizona measurements.

When diurnal measurements started, it had been raining for several hours. The rain persisted until around 18:00. Sunset on November 3 occurred at 17:45, and sunrise on November 4 occurred at 06:30. Measurements were taken every 15 min. Calibration measurements were taken every 6–8 h. These calibration measurements showed little change over the period of measurements. Figure 5 shows reference images from the diurnal measurements.

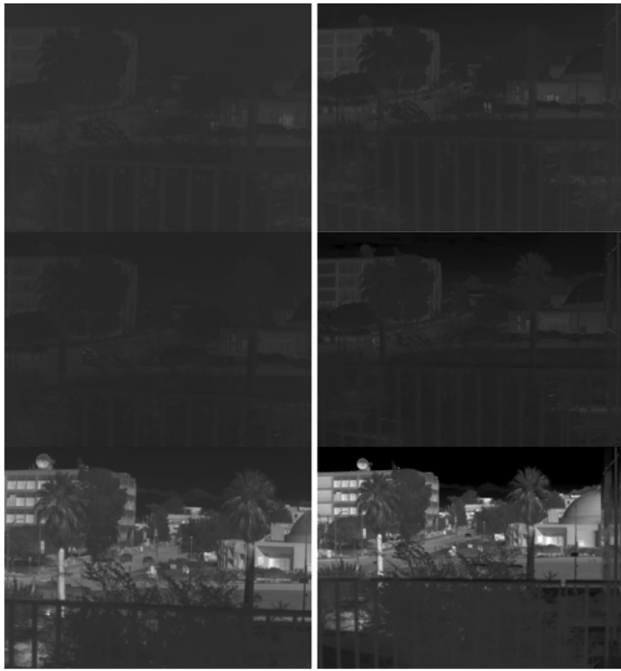


Fig. 5. Diurnal measurements in the MWIR (left) and LWIR (right) during the day when it was raining (top), during the night when it was overcast (middle), and during the day when it was clear and sunny (bottom). The highest scene variance is observed during the day, while the lowest is observed during the rain. The range of intensities shown in these images was fixed to the intensities that correspond to temperatures between 0 and 50°C. This figure shows the difficulty of navigating in a scene with reduced variance.

D. Measurements Taken in Colorado

Measurements were taken in Colorado—a cold, dry climate—from December 19 to December 25. Urban measurements were taken in Grand Junction, Colorado. The objects present in urban scenes included houses, roads, office buildings, and cars. Rural measurements were taken at Little Park Trailhead, Colorado. Objects in the rural measurements included scrub brush, small trees, rocks, and gravel. Snow measurements were taken in Grand Mesa, Colorado. The objects present in the snow measurements included rocks, trees, logs, and rocks, all covered by approximately 10 cm of snow. Calibration temperatures had to be lowered to a starting temperature of approximately 0°C and increased to a final temperature of 30°C. Table 3 lists the conditions during each Colorado measurement. Figure 6 shows reference images from the Colorado measurements, and Fig. 7 shows reference images from snow measurements. No nighttime measurements of snow were collected.

During diurnal measurements, sunset occurred at 17:15 on December 22 and sunrise occurred at 07:30 on December 23. The scene being imaged included a gravel yard, metal chairs, concrete, and wood. Unfortunately, the initial calibration data were unusable due to blackbody placement, so the calibration data from 22:00 were used for all measurements between 14:30 on the 22nd and 06:00 on the 23. This did not influence our measurements significantly because the calibration data showed very little change over several days.

4. RESULTS

Figure 8 shows the flux variance in equivalent blackbody temperature for the urban and rural scenes measured in Arizona and

Table 3. Description of Time, Location, and Weather Conditions of Measurements Taken in Colorado

Date	Time	Location	Purp.	Int. Time (μs)	Temperature	Hum.
Dec. 19	11:00–12:15	Grand Junction, CO	Urban day	1500	2.78–6.11°C	45%
Dec. 19	22:00–23:45	Grand Junction, CO	Urban night	5000	(−2.22)–(−1.67)°C	65%
Dec. 20	10:00–11:30	Little Park Trail, CO	Rural day	2000	0.563–4.45°C	40%
Dec. 20	22:30–23:15	Little Park Trail, CO	Rural night	4000	(−1.11)–(−1.67)°C	60%
Dec. 22–23	14:30–14:30	Grand Junction, CO	24 h	4000	(−2.22)–10.0°C	47%–68%
Dec. 25	13:30–15:00	Grand Mesa, CO	Snow	4000	(−3.34)	69%

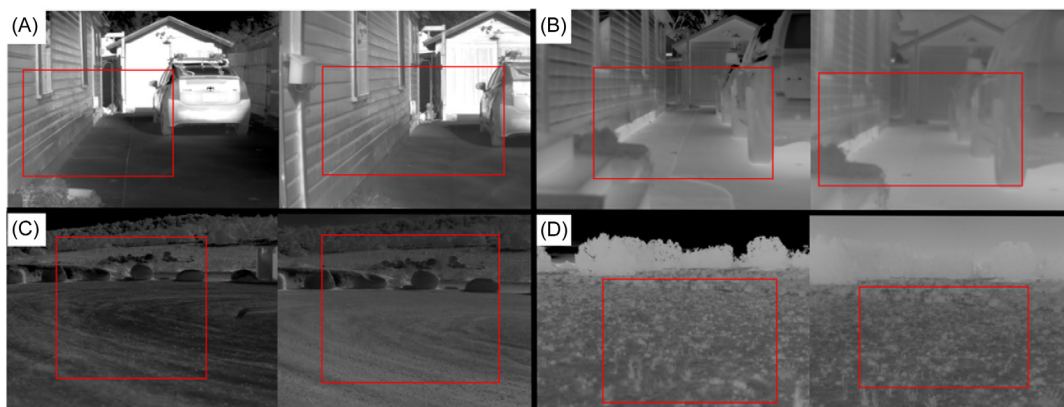


Fig. 6. Measurements taken in Colorado: daytime urban measurements in (A); nighttime urban measurements in (B); daytime rural measurements in (C); and nighttime rural measurements in (D). For each subfigure, the image on the left is of the LWIR and the image on the right is of the MWIR scene.

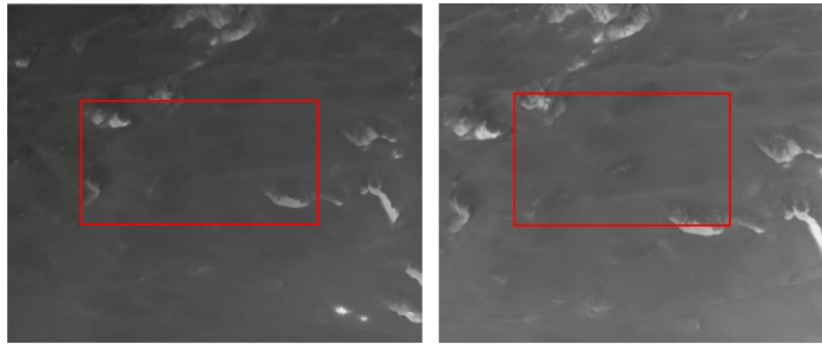


Fig. 7. Measurements taken of snow approximately 10 cm deep in the (left) LWIR and the (right) MWIR.

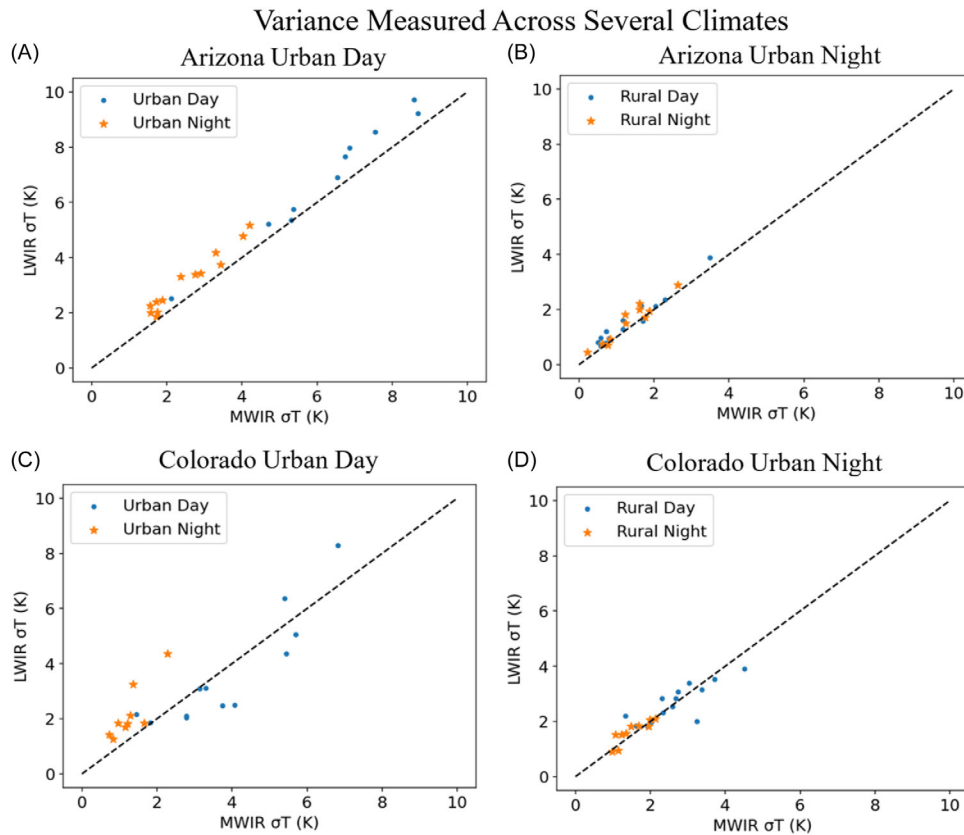


Fig. 8. Standard of deviation: (A) urban setting in Arizona, (B) rural setting in Arizona, (C) urban measurements in Colorado, and (D) rural measurements in Colorado.

Colorado. The standard deviation in matching subregions was used instead of the global SCT because the cameras had different FOVs.

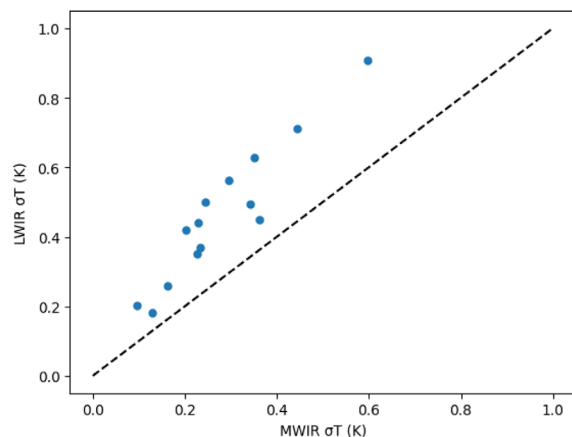
The urban measurements taken in Arizona, shown in Fig. 8(A), showed a strong linear relationship between the flux variance in the LWIR and MWIR, but with a higher offset in the LWIR, such that the flux variance was uniformly higher in the LWIR than in the MWIR. All but one of the daytime measurements had a higher flux variance than the nighttime measurements. The rural measurements taken in Arizona, shown in Fig. 8(B), also showed a strong linear relationship between the LWIR and MWIR flux variance and a higher offset in the LWIR, though not as strongly as in the urban measurements, with some measurements indicating a higher

flux variance in the MWIR. The difference between day and night measurements is also much reduced from the urban measurements. Table 4, subsection Arizona, summarizes these results.

The urban measurements taken in Colorado, shown in Fig. 8(C), showed a moderately strong linear relationship between the flux variance in the MWIR and LWIR. At night, the LWIR had a uniformly higher flux variance, while during the day, the MWIR tended to have a higher flux variance. Daytime measurements showed a higher flux variance than nighttime measurements. The rural measurements taken in Colorado, shown in Fig. 8(D), showed a stronger linear relationship in flux variance than the urban measurements did. The separation in flux variance in the MWIR and LWIR was approximately equal

Table 4. Average Standard Deviation in the LWIR and MWIR for Locations Measured

	LWIR Ave. Standard Deviation (K)	MWIR Ave. Standard Deviation (K)
Arizona		
Urban day	6.89	6.24
Urban night	3.16	2.56
Rural day	1.58	1.34
Rural night	1.54	1.31
Colorado		
Urban day	3.62	3.87
Urban night	2.18	1.27
Rural day	2.67	2.67
Rural night	1.62	1.50

**Fig. 9.** Standard of deviation for rural measurement taken in Colorado where snow covered the scene.

during the day and slightly higher in the LWIR at night. Table 4, subsection Colorado, summarizes these results.

Figure 9 shows the flux variance of the measurements taken in a snow-covered setting. Here, the overall flux variance was small. The variance in the two bands showed a linear relationship, but with an offset such that the variance in the LWIR was uniformly higher than that of the MWIR.

Figure 10(A) shows the results of the diurnal measurements taken in Arizona. The variance-reducing effect of rain on scene flux is shown while the sun is still up at the beginning of the

Table 5. Summary of Diurnal Measurements Including the Mean Standard Deviation, the Minimum Standard Deviation When the Minimum Was Measured, and the Maximum Standard Deviation When It Was Measured

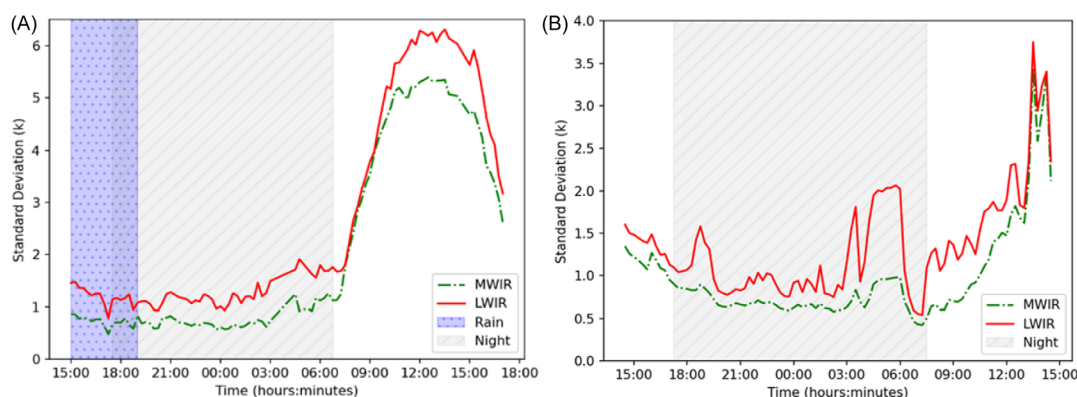
	LWIR Standard of Deviation	MWIR Standard of Deviation
Arizona		
Mean st. dev (K)	2.59	2.04
Minimum st. dev (K)	0.79 at 17:15	0.48 at 17:15
Maximum st. dev (K)	6.31 at 13:30	5.39 at 12:30
Colorado		
Mean st. dev (K)	1.36	0.97
Minimum st. dev (K)	0.54 at 07:15	0.42 at 07:15
Maximum st. dev (K)	3.75 at 13:30	3.44 at 13:30

measurements. The maximum flux variance occurred around noon, while the minimum flux variance occurred during the rain. The flux variance increased slightly overnight after the rain stopped, then increased rapidly after sunrise before falling again in the afternoon.

Figure 10(B) shows the results of the diurnal measurements taken in Colorado. The maximum flux variance observed occurred at 13:30, while the minimum variance occurred at 07:15, before sunrise. Some irregularities occurred during the diurnal measurements in Colorado, most noticeably between 04:00 and 06:00. It is unclear what caused this phenomenon, though it was likely due to manmade changes in the local environment. Table 5 summarizes the results of the diurnal measurements in both climates.

5. DISCUSSION

The data show that the flux variance in urban environments is greater than in rural ones. This is likely due to urban features having more varied emissivity than rural ones. Additionally, the flux variance is greater during the day than at night, likely due primarily to solar loading. Under a source of electromagnetic radiation (e.g., the sun), objects absorb energy and rise in temperature until they reach thermal equilibrium. Objects that are more absorptive to light absorb more energy and thus must reach a higher temperature to shed that energy via thermal radiation. Thus, the result of daylight is not simply hotter objects, but greater variation in object temperature, increasing flux

**Fig. 10.** Diurnal measurements taken in: (A) Arizona and (B) Colorado.

variance. Overnight, when there is no source of input radiation, objects release stored thermal energy as radiation and approach the background temperature, reducing flux variance. Similarly, the flux variance observed in the hot climate was larger than that in the cold climate due to the reduction in thermal energy present. When less energy is present, the effect of differences in absorptivity on temperature is reduced, and thus temperature variance and flux variance are reduced.

The diurnal measurements taken in Arizona indicate that rain reduces the flux variance present in a scene significantly, indicating that rain causes objects to release much of their stored thermal energy, also reducing flux variance. Finally, measurements in the snow show exceptionally little variance in the MWIR or LWIR, though there is more flux variance in the LWIR. The low flux variance in the snow is likely due to its uniformity in the scene, while the LWIR's increased relative variance is due to snow being highly absorptive in the thermal infrared and emitting photons based on its surface temperature [9,10].

In sum, the results indicate that the LWIR tends to have greater flux variance than the MWIR, except in cold, daylight conditions. This is likely due to the increased proportion of reflected light present in the MWIR [1]. More information about the reflective light present in the MWIR can be found in the work of Leslie *et al.* [11].

6. CONCLUSION

SCT, and the corresponding variance in equivalent blackbody temperature, is a key quantity that is used to determine performance in infrared systems for applications like pilotage and targeting. In applications like pilotage, where situational awareness is desired, high SCTs are desirable. In applications like targeting, where high contrast between targets and the background and low clutter are desired, low SCTs are desirable.

As a measure of the variance in scene temperature, the standard deviation of flux in a scene is closely related to SCT. The standard deviation of scene temperatures was measured in the MWIR and LWIR in hot and cold climates, during the day

and night, in both urban and rural environments. The results of these measurements indicate that the variance in the LWIR tends to be larger than the MWIR except in cold, daylight conditions when snow is not present. Thus, for applications like pilotage systems, LWIR may provide greater situational awareness than MWIR, while for applications like targeting, MWIR may provide higher target contrasts and reduced clutter.

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Data availability. The data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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